

Bagus Satrio Wicaksono

Electronic Engineer

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SUMMARY

Electronic Engineer with hands-on experience as a Manufacturing Engineer at Polytron Indonesia, specializing in Industrial Internet of Things (IIoT), Embedded Systems, Machine Learning, Computer Vision, and 3D Design. Experienced in DSP, bare-metal programming, and embedded software development for manufacturing applications. Contributed to AI and IoT based systems across industrial and research projects, including robotics, web-based platforms, and rehabilitation systems with pose estimation. Proficient in C/C++, Python, TensorFlow, MATLAB, STM32, ARM Cortex-M4, Arduino, CAD Design, and PCB Design. A fast learner and detail-oriented engineer with strong problem-solving skills and a passion for developing innovative technology solutions.

EDUCATION

Satya Wacana Christian University | Salatiga, Indonesia

Bachelor of Engineering (August 2021 – November 2025)

- **GPA** (3.80/4.00)
- **Relevant Courses:** Electronic Engineering, Robotics, Embedded Software Engineering, Quality Control, Computer Vision
- **Thesis Title:** Disease Detection on Ginger Emprit Leaves Using the YOLO Algorithm

PROFESSIONAL EXPERIENCE

PT Hartono Istana Teknologi (Polytron) | Kudus, Indonesia

Fulltime Manufacturing Engineer (January 2026 – Present)

- Analyzed workflows with cross functional team productions to identify process improvement opportunities optimize efficiency.
- Designed and implemented Industry 4.0 solutions by IIoT and automation technologies into manufacturing processes.
- Performed cost analysis and feasibility studies for automation and process improvement initiatives.

[Intern Research Function Checker Module Audio](#) (September 2024 – March 2025)

- Developed a function checker using bare-metal programming, achieving 94% cost reduction.
- Collaboration with team to developing electronic custom circuit, enhancing 62% faster testing time and data management.

PT Dicoding Akademi Indonesia | Bandung, Indonesia

[Freelance External Academy Code Reviewer at Machine Learning](#) (August 2025 – Present)

- Evaluated hundreds of code submissions to ensure quality and scalability for developers in the Dicoding.
- Provided detailed feedback that improved learners' coding proficiency, contributing to higher completion rates.

Indonesian Association of Health Service Engineering | Jakarta, Indonesia

[Intern Information Technology Engineer](#) (September 2025 – October 2025)

- Designed a 3D model of an inpatient room using 3DS Max to visualize the floor plan, optimizing spatial efficiency.
- Designed a 3D model of a hall room using Blender, enhancing spatial planning for the INAHEF Southeast Asia.

PT Full Drone Solution | Yogyakarta, Indonesia

[Intern Electronic Engineer](#) (February 2024 – June 2024)

- Study Program MSIB 6, gaining hands-on experience in electronics and control systems to expertise in UAV technology.
- Developed technical through collaborative projects, enhancing professional readiness to understanding UAV application.

Telekomunikasi Tunas Harapan Vocational High School. | Semarang Recency, Indonesia

Freelance Robotic Mentor (November 2022 – November 2023)

- Mentored in developing project and paper, win 3rd place in the Semarang Regency Innovation Contest.
- Taught robotics classes, delivering electronics, sensors, and robotics, to strengthen students' technical competencies.

Senior High School 1 Salatiga | Salatiga, Indonesia

Freelance Robotic Mentor (January 2023 – June 2024)

- Mentored through project-based learning and scientific writing, got 5th place in the Salatiga Innovation Contest.
- Delivering basic electronics, C programming, and 3D design, fostering innovation and interest in robotics.

Satya Wacana Christian University | Salatiga, Indonesia

Assistant Lecture of Digital System (December 2023 – June 2024)

- Assisted the lecturer by preparing teaching materials and assessments, delivery of concepts in digital logic and circuit design.
- Guided class discussions and practical sessions, enhancing their understanding of digital logic implementation.

Assistant Lecture of Basic Electrical Engineering (January 2023 – December 2023)

- Assisted the lecturer by preparing teaching materials and assessments, delivery of concepts in electronic law and component.
- Supervised practical sessions, guiding through hands-on experiments to strengthen their technical comprehension.

ORGANIZATIONAL EXPERIENCE

AIIESEC in Satya Wacana Christian University | Salatiga, Indonesia

Local Committee Vice President of Operation (February 2024 – February 2025)

- Led the AFL program, engaging 115 youth participants, coordinating 35+ sessions, achieving 9.25/10 NPS.
- Directed marketing and operational strategies, generating over 11.2 million IDR in program revenue.

Manager of Learning and Development (March 2023 – February 2024)

- Designing and facilitating structured learning programs that enhanced members' organizational management skills.
- Implement organizational strategies, strengthening operational effectiveness with corporate-style management practices.

Robotic Research Center | Salatiga, Indonesia

Head of Robotics Rescue Division (June 2024 – June 2023)

- Developed movement algorithms for hexapod robot using 3-DOF inverse kinematics and achieving gait patterns efficiently.
- Collaborated with a team to design a functional robot, advancing to the national Top 16 in the KRI competition.

Embedded System Programmer (February 2022 - June 2024)

- Programmed microcontroller-to integrated multiple sensors to enable real-time data acquisition and robot operation.
- Developed communication protocols and control algorithms, enhancing system reliability and coordination in robot performance.

HONORS, AWARDS, and SKILLS

- **Competitions:**
 1. [1st Best Group Samsung Innovation Campus Indonesia Batch Seven \(2026\)](#).
 2. [1st Renewable Energy Write Competition \(2025\)](#).
 3. [Finalist KRI \(Kontes Robot Indonesia\) National \(2024\)](#).
 4. [Finalist KRI \(Kontes Robot Indonesia\) National \(2022\)](#).
 5. [2nd place in the Student Creativity Program, Satya Wacana Christian University \(2022\)](#).
 6. [1st of the technology innovation competition in the era of Society 5.0 \(2022\)](#).
- **Boothcamp:**
 1. Full stack Engineer Cohort at Coding Camp Powered by DBS Foundation (2026).
 2. [AI and IoT Samsung Innovation Campus Indonesia Batch Seven \(2026\)](#).
 3. [Machine Learning Cohort at Coding Camp Powered by DBS Foundation \(2025\)](#).
- **Certifications:**
 1. [DSP From Ground Up™ on ARM Processor \(2025\)](#).
 2. [Advanced Embedded Systems Bare Metal Programming Ground Up™ \(2024\)](#).
 3. [Embedded System Bare-Metal Programming Ground Up™ \(2024\)](#).
 4. [YOLOv8 & YOLO11: Custom Object Detection & Web Apps \(2025\)](#).
- **Volunteering:**
 1. [Indonesia Social Action Ambassador \(2024\)](#).
 2. [Pancasila Ambassador of Indonesia's Paskibraka \(2023\)](#).
- **Published:**
 1. [A Training in Making Technology Innovation for Students of SMA Negeri 1 Salatiga \(2024\)](#).
- **Projects:**
 1. [Monitoring HR, Humidity, SpO2, GSR using Modbus TCP, MySQL, telegram and website \(2026\)](#).
 2. [PhysioTrack: Smart Assistance for Stroke Therapy \(2026\)](#).
 3. Designing Synchronization and Protection System with Monitoring for STM32 Microgrid (2025).
 4. Emergency Button for Hospital Rooms (2025).
 5. Disease Detection on Ginger Emprit Leaves Using the YOLO Algorithm (2025).
 6. [Monitoring and Prediction of Papaya Plants Based on IoT and Machine Learning \(2025\)](#).
 7. [JIG Monitor and Function Checker Module Audio Using STM32 \(2024\)](#).
 8. [Hexapod Robot with Six Legs \(2024\)](#).
 9. Quadruped Robot with Four Legs (2023).
- **Technical Skills:** Cortex M4, Arduino, Computer Vision, PCB Design, 3D Design, Python, MATLAB, Microsoft Office.
- **Interpersonal Skills:** Critical Thinking, Teamwork, Time Management, Problem Solving, Public Speaking.
- **Languages:** English (B1), Japanese (A2), Indonesian (Native), Javanese (Native).